

Deep Learning for Audio

Lecture 4

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AI Masters

2025

Outline

1. Keyword spotting (KWS): task overview
2. KWS: history & models
3. KWS: Cascade Systems
4. KWS on practice

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Key Word Spotting task

Where:

1. When user is initiating interaction with device?
2. Searching through audio stream by keyword

Other names:

- ▶ Phrase Spotting
- ▶ Wake Word Detection
- ▶ Hotword Detection
- ▶ Trigger Word Detection

KWS: introduction

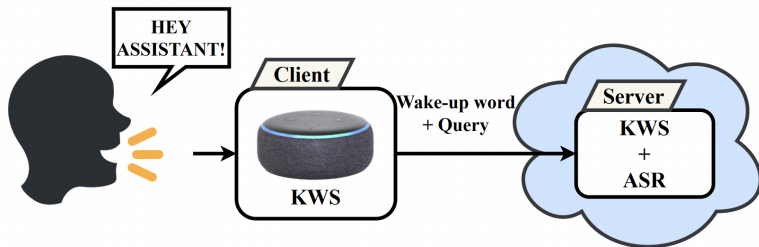
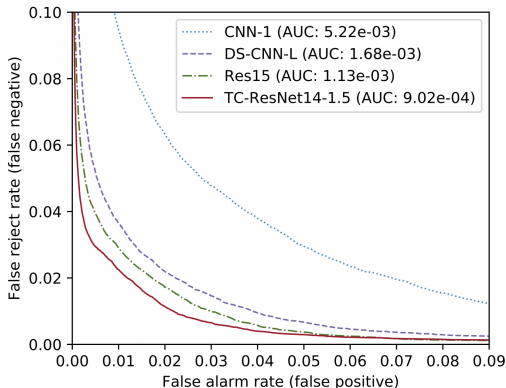


Figure: Typical voice assistant client-server framework

What do we need? Lightweight robust ASR model

KWS: quality



- ▶ False alarm rate (false positive rate): $FAR = \frac{FP}{FP+TN}$
- ▶ False Reject Rate (false negative rate): $FRR = \frac{FN}{FN+TP}$
- ▶ FA/FR per hour, lower curves are better

Choi et al., *Temporal Convolution for Real-time Keyword Spotting on Mobile Devices*, 2019

KWS on Google Speech Commands Leaderboard

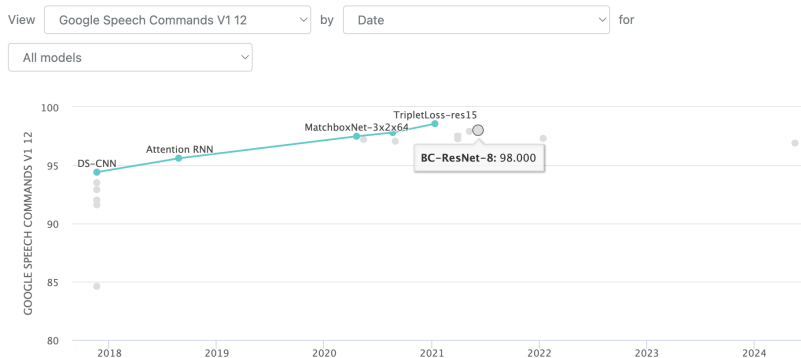
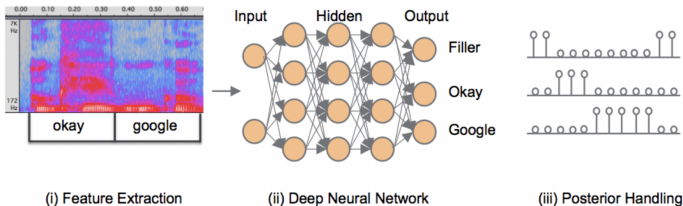


Figure: v1: 65,000 one-second long utterances of 30 short words, by thousands of different people;
v2: 105,829 utterances of 35 words

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DNN based KWS



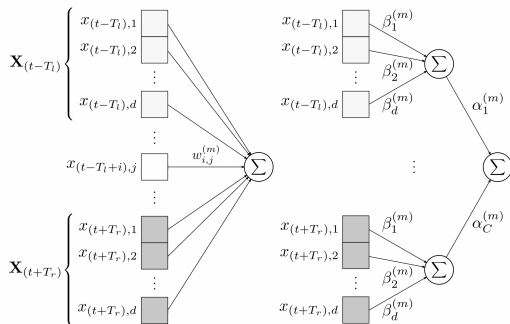
- Feature extraction: time windows w_{smooth} : 10 future frames + 30 frames in the past
- Raw posteriors from DNN are noisy, so smooth them over w_{smooth}

$$p'_{ij} = \frac{1}{j - h_{smooth} + 1} \sum_{k=h_{smooth}} p_{ij},$$

$h_{smooth} = \max\{1, j - w_{smooth} + 1\}$ – the index of the first frame within the smoothing window

Chen et al., *Small-footprint keyword spotting using deep neural networks*, IEEE International Conference on Acoustics, Speech and Signal Processing, 2014

Single Value Decomposition Filter (SVDF)



Before: $a_t^{(m)} = f \left(\sum_{i=0}^{C-1} \sum_{j=1}^d w_{i,j}^{(m)} x_{(t-T_l+i),j} \right)$

After:

$$a_t^{(m)} \approx f \left(\sum_{i=0}^{C-1} \alpha_i^{(m)} \sum_{j=1}^d \beta_j^{(m)} x_{(t-T_l+i),j} \right), \quad w_{i,j}^{(m)} \approx \alpha_i^{(m)} \beta_j^{(m)}$$

Nakkiran, Preetum et al., Compressing deep neural networks using a rank-constrained topology, INTERSPEECH 2015

CNN based KWS

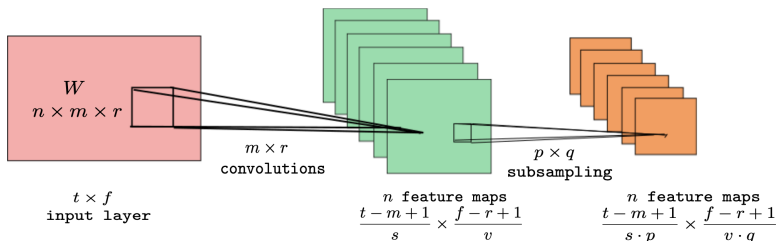
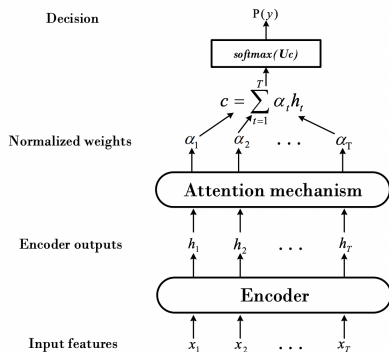


Figure: CNN architecture

- ▶ Input signal $V \in \mathbb{R}^{t \times f}$, t and f – input feature dimension in time and frequency.
- ▶ Weight matrix $W \in \mathbb{R}^{(m \times r) \times n}$ is convolved with input V
- ▶ Filter of size $m \times r$, stride by s in time and v in frequency
- ▶ Pooling size $p \times q$

Sainath et al., *Convolutional neural networks for small-footprint keyword spotting, INTERSPEECH, 2015*

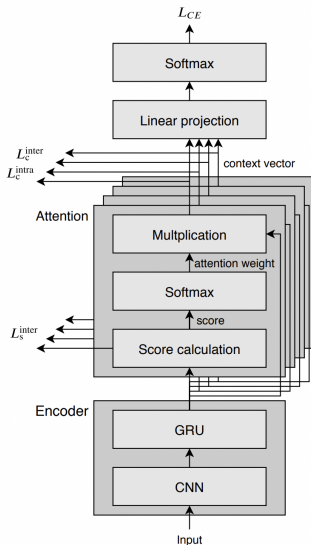
Attention-based KWS



► Encoder:
RNN/GRU/LSTM/CRNN
 $\mathbf{h} = \text{Encoder}(\mathbf{x})$

► Soft attention:
$$e_t = v^T \tanh(\mathbf{W}\mathbf{h}_t + \mathbf{b}).$$
$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^T \exp(e_j)}.$$
$$\mathbf{c} = \sum_{t=1}^T \alpha_t \mathbf{h}_t.$$

Multihead attention based KWS



- Retrieves richer information than a single-head attention which only summarizes the whole sequence into one context vector
- The diversity is not guaranteed by its natural form → use Orthogonality regularization

Multihead attention based KWS: orthogonality

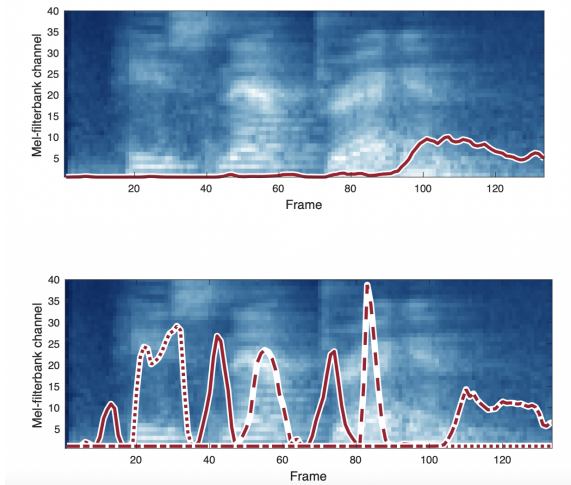


Figure: Attention weights overlaid on Mel-spectrogram with the configurations of single head attention (top), 4-head attention with regularization (bottom)

Multihead attention based KWS: orthogonality

- ▶ Context matrix and the score matrix from normalized vectors (H – number of attention heads):

$$\mathbf{C}^{(n)} = [\bar{\mathbf{c}}_1^{(n)}, \bar{\mathbf{c}}_2^{(n)}, \dots, \bar{\mathbf{c}}_H^{(n)}], \quad \mathbf{E}^{(n)} = [\bar{\mathbf{e}}_1^{(n)}, \bar{\mathbf{e}}_2^{(n)}, \dots, \bar{\mathbf{e}}_H^{(n)}]$$

- ▶ Inter-head orthogonality regularization

$$\mathcal{L}_c^{\text{inter}} = \frac{1}{N_P} \sum_{n=1}^N \frac{y^{(n)}}{H(H-1)} \|\mathbf{C}^{(n)\text{T}} \mathbf{C}^{(n)} - \mathbf{I}_H\|_F^2$$

$$\mathcal{L}_s^{\text{inter}} = \frac{1}{N_P} \sum_{n=1}^N \frac{y^{(n)}}{H(H-1)} \|\mathbf{E}^{(n)\text{T}} \mathbf{E}^{(n)} - \mathbf{I}_H\|_F^2$$

- ▶ Intra-head non-orthogonality regularization

$$\mathcal{L}_c^{\text{intra}} = \frac{1}{H} \sum_{i=1}^H \frac{1}{N_P(N_P-1)} \left\| \mathbf{Y}(\tilde{\mathbf{C}}_i^{\text{T}} \tilde{\mathbf{C}}_i - \mathbf{I}_N) \mathbf{Y} \right\|_F^2$$

- ▶ Cross entropy loss with the regularization terms

$$\theta^* = \arg \min \{ \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_c^{\text{inter}} - \lambda_2 \mathcal{L}_c^{\text{intra}} + \lambda_3 \mathcal{L}_s^{\text{inter}} \}$$

Convolutions in audio

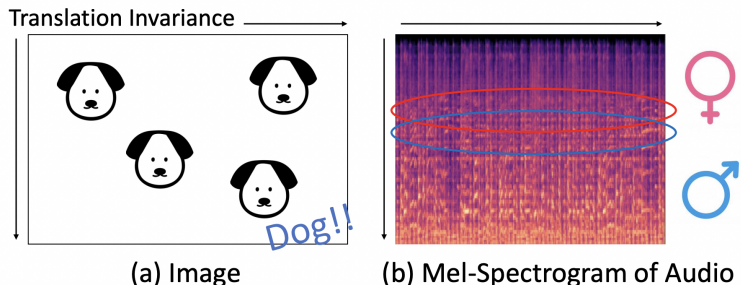


Figure: 2D convolution on image and audio input: unlike image processing, the feature in different audio frequency bands has different information.

MatchboxNet: KWS with 1D convolution

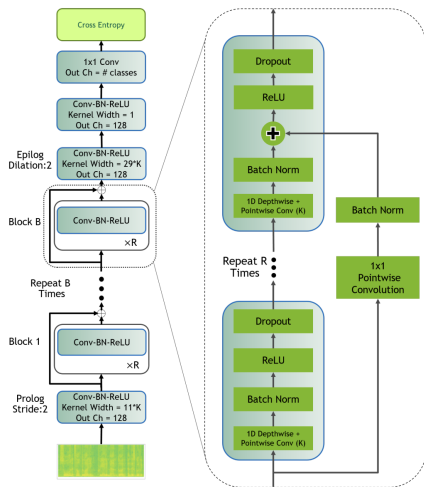


Figure: MatchboxNet builds on the QuartzNet architecture

Majumdar, Ginsburg, MatchboxNet: 1D Time-Channel Separable Convolutional Neural Network Architecture for Speech Commands Recognition, NVIDIA, 2020

Subspectral Normalization (SSN)

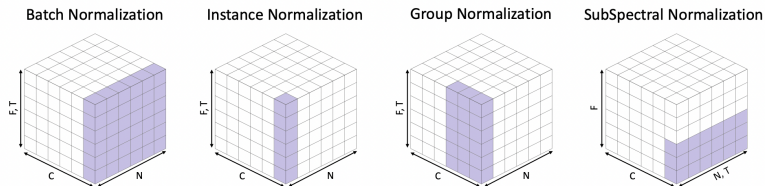
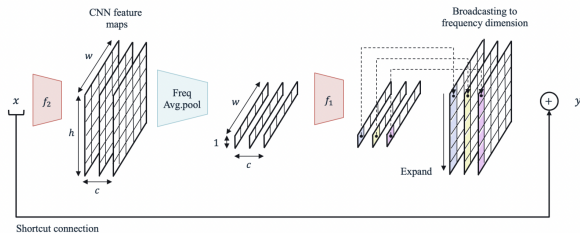


Figure: Normalization methods on Frequency-Time audio input, N – batch axis, C – channels, F – frequency, T – time axis. SSN splits the frequency dimension into multiple sub-bands and normalizes each group

Broadcasted Residual Learning

Broadcasted Residual Learning



Normal block

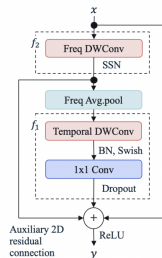
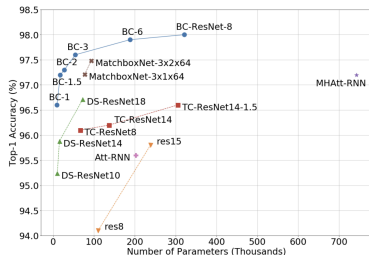
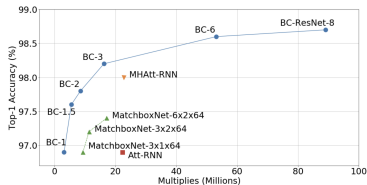


Figure: Left, Broadcasted Residual Learning $y = x + BC(f_1((f_2(x))))$, number of channels c .
Right, BCResBlock $y = x + f_2(x) + BC(f_1(avgpool(f_2(x))))$.

Broadcasted (BC) Residual Learning: metrics



Model Size vs. Google speech
command dataset v1 Test
Accuracy



MACs vs. Google speech
command dataset v2 Accuracy

Keyword Transformer based KWS

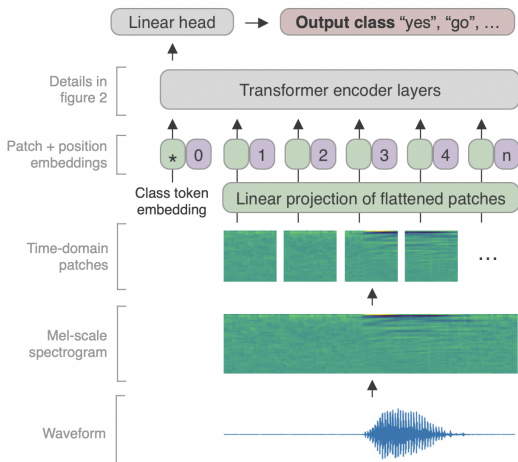


Figure: Keyword Transformer architecture

Berg et al., Keyword Transformer: A Self-Attention Model for Keyword Spotting. Interspeech, 2021

KWS: summary

KWS Idea	Year/Company	Params	Accuracy, V1 dataset
DNN	2014, Google	~224k	91.2
CNN	2015, Google	20-70k	95.4
Attention	2018, Xiaomi	~84k	95.6
Multihead attention	2019, Qualcomm	743k	97.2
Single Value Decomposition Filter	2015/2019, Google	40-700k	96.3
Broadcasted Residual	2021, Qualcomm	9.2k	96.6
MatchboxNet	2020, Nvidia	77-93k	97.4
Transformer	2021, Arm ML Research Lab	607-5361k*	97.5

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KWS: Google Cascade Systems

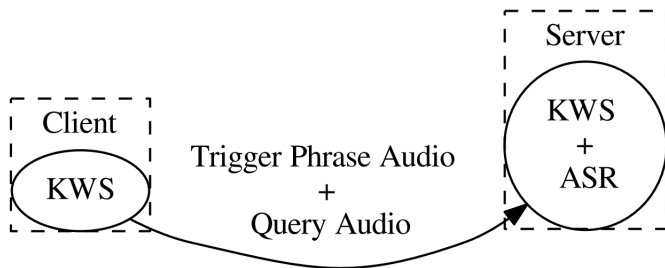


Figure: Client-Server keyword spotting system: KWS on device + KWS on server for validation (with ASR)

KWS: Google Cascade Systems [2]

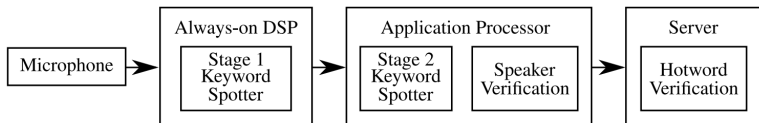
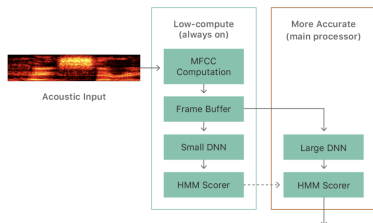
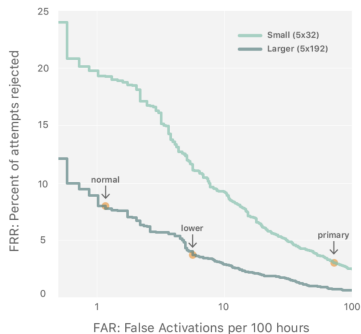


Figure: Cascade keyword spotting architecture: small footprint detector + more accurate detector + speaker verification on mobile devices

KWS: Apple Cascade Systems, 2017



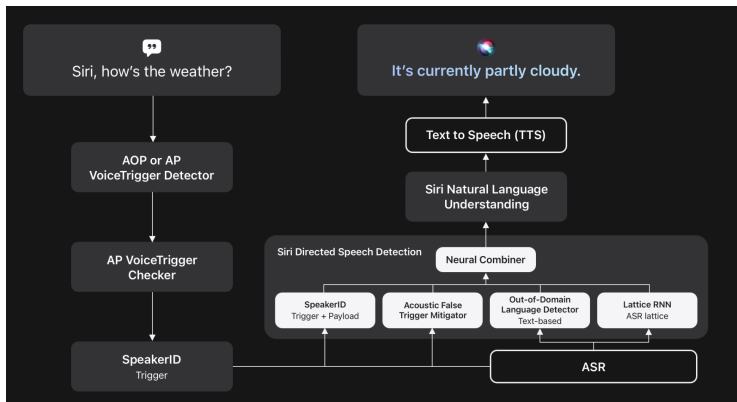
Hey Siri: Two-pass detection



Detector accuracy, Trade-offs

Siri Team, Hey Siri: An On-device DNN-powered Voice Trigger for Apple's Personal Assistant, Apple, 2017

KWS: Apple Cascade Systems, 2023

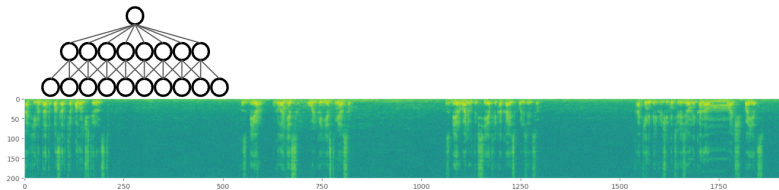
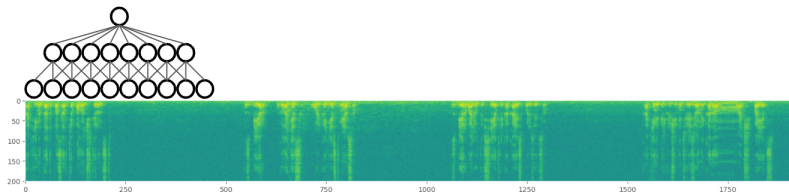


1. Streaming on the Always On Processor (AOP)
2. Audio that may contain the keywords goes to a high-precision voice trigger checker on the Application Processor (AP)
3. For personal devices, like iPhone: speaker identification
4. ASR on server

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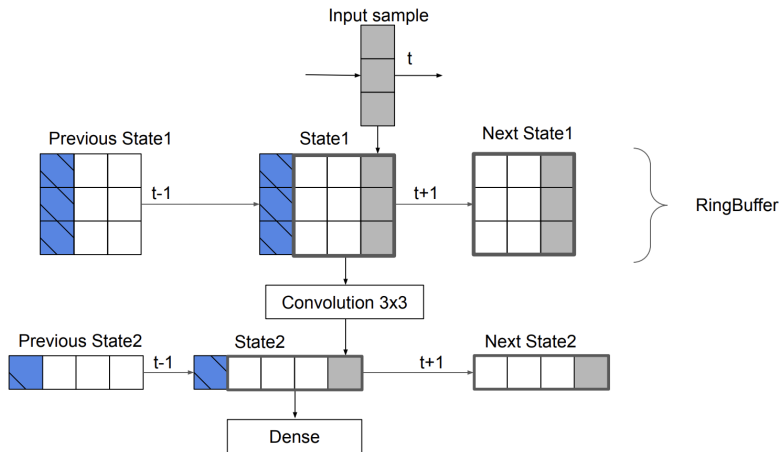
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KWS: Streaming Detection

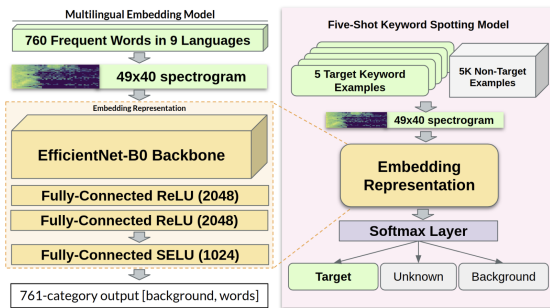


Want to cache results of convs

KWS: Google Streaming Detection



KWS: transfer learning



(a) Multilingual embedding model

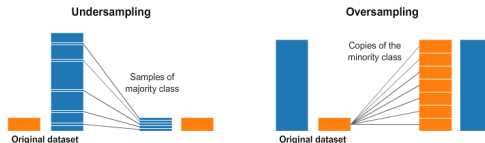
(b) 5-shot keyword spotting

- ▶ Embedder from classifier on 760 keywords totaling 1.4M samples in nine languages
- ▶ To train a new KWS model, can fine-tune a 3-category classifier using just 5 target examples and 128 non-target samples from a precomputed “unknown” keyword bank

Mazumder1 et al., Few-Shot Keyword Spotting in Any Language,
Harvard+Coqui+Google, 2021

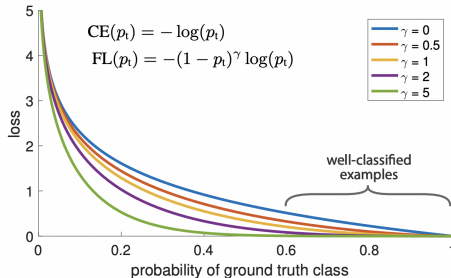
Class imbalance

► Oversampling



► Class weighted Cross-entropy (CE)

► Focal Loss



Lin, Goyal et al., *Focal Loss for Dense Object Detection*, in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, 2020

Robustness

- ▶ Multitask learning: Train ASR and KWS simultaneously

$$\mathcal{L} = \gamma \mathcal{L}^{(1)} + (1 - \gamma) \mathcal{L}^{(2)}, \quad 0 \leq \gamma \leq 1$$

- ▶ Augmentations
 - ▶ Pitch and Tempo
 - ▶ Time shifts
 - ▶ Resampling
 - ▶ Environment Simulation
 - ▶ Background noise
 - ▶ Generate new data (Text-to-Speech)
- ▶ Hard negative mining
- ▶ Federated Learning
- ▶ Clean data

Model optimization

- ▶ **Graph Optimization**

RepCNN: Micro-sized, Mighty Models for Wakeword Detection (Apple, 2024)

- ▶ **Quantization**

Low-bit quantization and quantization-aware training for small-footprint keyword spotting (Amazon, 2019)

- ▶ **Pruning**

Accurate and Structured Pruning for Efficient Automatic Speech Recognition (Microsoft, 2023)

- ▶ **Parameter Sharing**

Leveraging Adapter for Parameter-Efficient ASR Encoder (Qualcomm, 2024)

- ▶ **Distillation**

Noisy student-teacher training for robust keyword spotting (Google, 2021)